View results

Respondent

52 Anonymous

274:31

Time to complete

A. General information about the respondant

1. * Country

Libya

2. * Name of organization(s) or entity(s) responsible for the preparation of the report

This report has been prepared by the Libyan National Focal Point and the Libyan National Committee for Open Science

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

A specific website for the National Committee for Open Science has not yet been established. However, in the near future, we will have a dedicated website for the National Committee. The National Committee represents all organizations and ministries related to open science.

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

7. * Full name of officially designated contact person(s) who completed this survey

Dr. Abdolbast Greede,

8. * Position of officially designated contact person(s) who completed this survey

Libyan National Focal Point

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

11. Brief description of the consultation process established for the preparation of the report

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

To complete this report, the National Focal Point and the Libyan National Committee for Open Science, with direct support from the Minister of Higher Education, held consultations with the following entities: Libyan Open University, Libyan Authority for Scientific Research, Ministry of Education, Ministry of Culture and Knowledge Development, and the National Council for Social and Economic Development.

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

Ministry of Higher Education and Scientific Research (https://mhesr.gov.ly/), Libyan Authority for Scientific Research (https://www.aonsrt.ly/), Ministry of Education (https://moe.gov.ly/), Ministry of Culture and Knowledge Development (https://www.culture.gov.ly/).

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

- [x] Public
- [ ] Private
- [ ] Civil Society
- [ ] Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

- (●) Yes
- ( ) No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

Ministry of Higher Education and Scientific Research, Libyan Authority for Scientific Research, Ministry of Education, and Ministry of Culture and Knowledge Development

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

- [x] Yes
- [ ] No

19. 1.2 cont. **If yes,** which language(s)?

Arabic

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? *(ref.: (i) 16.f, 16.g, 16.h)*

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

- [ ] Yes
- [ ] No

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

21. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

    [i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

    - ○ Yes
    - ◉ No

22. 2.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

    Although Libya has not yet developed a national policy, strategy, or action plan dedicated to science, technology, and innovation, the country has formulated a national strategy for the higher education sector. In addition, several organizations working in the field of science and technology have developed national strategies that contribute to the advancement of these areas. The National Committee for Open Science, which includes representatives from key sectors involved in promoting science, technology, and innovation, recognizes that these fields are essential drivers of sustainable development. Therefore, the National Committee for Open Science will advocate for the development of a national framework for science, technology, and innovation in Libya that is aligned with national priorities and contributes to sustainable development.

23. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- Yes
- Partly
- No

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

24. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

25. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

26. 3.3 Are there national research and education networks (NRENs)[i] active in the country? *(ref.: (iii) 18.c)*

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

- ○ Yes
- ◉ No

27. 3.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

Libya does not have a specific body named the "National Research and Education Network (NREN)." In fact, research and education institutions in Libya operate under three different authorities, each with distinct responsibilities: 1. Ministry of Education – Responsible for compulsory education. 2. Ministry of Higher Education and Scientific Research – Comprising two separate entities: - Libyan Authority for Scientific Research, which oversees 45 research centers. - Libyan Universities**, which consist of 27 universities. 3. Ministry of Technical and Vocational Education– Oversees technical and vocational education institutions. While there is no centralized NREN, these institutions contribute to research and education in Libya through their respective frameworks. However, the Libyan National Committee for Open Science understands the importance of an NREN for the country; therefore, the Committee will work with Libyan governmental institutions to highlight the need to establish one in Libya.

28. 3.4 Are there national or institutional open science infrastructures[i] in your country? *(ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)*

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- (•) Yes
- ( ) Under development
- ( ) No

29. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- [x] federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- [x] community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- [ ] platforms for exchanges and co-creation of knowledge between scientists and society
- [ ] community-based monitoring and information systems to complement national, regional and global data and information systems

30. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

The Libyan education system is built on the principles of open science, focusing on enhancing access to education and ensuring open access to scientific publishing. For example, the Libyan Ministry of Education succeeded in integrating all scientific journals affiliated with universities and research centers into a single platform "https://lsjp.epc.ly/ar/Journal". It stipulated that all these journals be open access, ensuring that scientific knowledge is freely available to researchers, students, and the wider community. Moreover, Libya was one of the first countries to establish an open university in 1987 "https://ou.edu.ly/", pioneering the concept of open education and promoting higher education for all segments of society. This open education model has proven effective in expanding learning opportunities throughout Libya, especially in areas with limited access to traditional educational institutions. Looking ahead, Libya aims to strengthen its commitment to open science through the following initiatives: - Expanding digital platforms to facilitate the sharing of research outputs and datasets. - Encouraging the integration of open science policies into higher education institutions and research centers. - Developing national guidelines to support open publishing and open data repositories. - Implementing training programs to build researchers' capacities in open science methodologies. - Enhancing international collaboration to align Libya's open science initiatives with global standards. While Libya has already taken significant steps in promoting open science principles, we recognize the need to further enhance our efforts by implementing innovative policies and developing infrastructure that ensures sustainable openness, accessibility, and collaboration in scientific research.

31. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

- ○ Yes
- ○ No

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY

AND CAPACITY BUILDING FOR OPEN SCIENCE

32. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

- [ ] Yes
- [x] No

33. 4.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

> No, systematic capacity building on open science has not yet taken place in Libyan higher education and research performing institutions. Although these institutions engage in various open science activities such as knowledge sharing and maintaining open access journals, they currently lack a formal and structured program dedicated exclusively to building capacity in open science. While foundational efforts exist, structured capacity building remains an area for further development.

34. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

- [ ] Yes
- [ ] No

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

35. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? *(ref.: (v) 20.b, 20c, 20i)*

- (●) Yes
- ( ) No

36. 5.1 cont. **If yes**, please provide more information and links as relevant.

Yes, initiatives are expected to be developed as part of the newly established National Committee for Open Science. With the recent formation of the National Committee for Open Science, Libya is now in a stronger position to develop policies and initiatives that align research assessment and evaluation processes with open science principles and values. The committee is expected to propose and implement various reforms that encourage open-access publishing, and transparent research practices across academic and research institutions. To achieve this goal, the committee aims to: • Review and modernize existing research evaluation frameworks to include open science indicators such as open-access publications, data sharing, and community engagement. • Encourage universities and research centers to integrate open science policies into their institutional assessment and funding criteria. • Promote the adoption of FAIR principles (Findable, Accessible, Interoperable, and Reusable) for research data management. • Develop national guidelines and policies to incentivize researchers to adopt open science practices and ensure their contributions to open knowledge are recognized and rewarded. • Engage with international open science networks to align Libya’s research evaluation standards with global best practices. Although these initiatives are still in the early stages, the National Committee for Open Science is fully committed to ensuring that open science principles are embedded in the research assessment and evaluation framework by 2025. By doing so, Libya aims to increase research transparency, enhance collaboration, and ensure broader access to scientific knowledge for both academic and non-academic communities.

37. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

- Yes
- No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

38. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? *(ref.: (vi) 21)*

- YES, at the national level
- YES, at the institutional level
- YES, at both the national and institutional levels
- No

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

39. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

At this stage, Libya has not yet established formal international scientific collaborations specifically focused on open science. However, the National Committee for Open Science is actively working toward the development of open science projects that will encourage international partnerships and knowledge exchange.

40. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

- ○ Yes
- ○ No

Part 8. GENERAL CONSIDERATIONS

41. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

- ◉ Yes
- ○ No

42. 8.1 cont. **If yes**, please provide more information.

Yes, several factors have contributed to the delay in implementing UNESCO recommendation, which can be summarized as follows: - Delay in appointing a National Focal Point:The official designation was only made in November 2024. - Lack of dedicated funding: No funds have been allocated for promoting and implementing the 2021 UNESCO Recommendation on Open Science. - Institutional and administrative challenges:Coordinating efforts among different ministries and research institutions has proven difficult. - Limited awareness and promotion: Delays in the formal dissemination of the Open Science Recommendation have hindered widespread awareness among key stakeholders. - Absence of a dedicated national policy or strategy: There is no established policy or strategy specifically for Open Science, which requires significant time for development and approval. Despite the obstacles mentioned above, we strongly believe that the promotion of Open Science and the implementation of UNESCO's recommendations will accelerate now that a National Focal Point has been appointed and the National Committee for Open Science has been established. To ensure that the National Committee for Open Science achieves its goals, it includes senior officials from key stakeholders, including: - Deputy Minister, Ministry of Education - Deputy Minister, Ministry of Technical and Vocational Education - President of the Libyan Open University - Deputy of the National Council for Social and Economic Development - General Manager of the Libyan Authority for Scientific Research - Head of the National Program for Enhancing the International Competitiveness of Libyan Universities - Head of the Consultant Committee at the Ministry of Higher Education and Scientific Research - Head of Affairs Office at the Ministry of Higher Education and Scientific Research - Head of the Legal Office at the Ministry of Higher Education and Scientific Research Libya is fully committed to aligning with UNESCO's vision, recognizing the immense value and benefits that Open Science brings to both the Libyan academic and non-academic communities. Moving forward, we are eager to accelerate our efforts in collaboration with UNESCO and international partners, ensuring that Open Science principles are embedded within Libya's higher education, research, and knowledge-sharing ecosystem.

43. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org